



Southeast University, Bangladesh

School of Science and Engineering

Department of Computer Science and Engineering

Lab Report

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Lab Report: Linux Basics

1. Objectives

The main objectives of this lab are:

- To understand the basic concepts of the Linux operating system.
 - To learn how to navigate through the Linux file system using the terminal.
 - To display the current working directory and list directory contents.
 - To create, remove, rename, and manage directories using Linux commands.
 - To understand the hierarchical structure of the Linux file system.
 - To gain practical experience with basic Linux terminal commands.
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2. Introduction

Linux is an open-source operating system widely used in servers, personal computers, cloud computing, networking, and cybersecurity. One of the most important features of Linux is its command-line interface (CLI), which allows users to interact with the operating system by entering commands in a terminal.

Linux provides many commands for managing files and directories. Basic commands such as `pwd`, `ls`, `cd`, `mkdir`, `rmdir`, `rm`, and `mv` are commonly used for navigating and managing the file system.

In this experiment, several fundamental Linux commands are studied and implemented. These commands help users understand the directory structure and perform basic file-system operations efficiently.

3. Implementation / Theory

3.1 `pwd` – Print Working Directory

The `pwd` command displays the absolute path of the current working directory.

Syntax:

```
pwd
```

Example:

```
$ pwd  
/home/user
```

Here, /home/user represents the current working directory.

3.2 ls – List Directory Contents

The ls command is used to display the files and directories available in the current directory.

Syntax:

```
ls
```

Example:

```
$ ls  
Documents Downloads Music Pictures
```

3.3 ls -p – Identify Directories

The ls -p command lists directory contents and adds a / after directory names. This makes it easier to distinguish directories from regular files.

Syntax:

```
ls -p
```

Example:

```
$ ls -p  
Documents/ Downloads/ file.txt
```

Here, Documents/ and Downloads/ are directories, while file.txt is a regular file.

3.4 cd <Foldername> – Change Directory

The cd command is used to move from the current directory to another directory.

Syntax:

```
cd <Foldername>
```

Example:

```
$ cd Documents
```

This command changes the current working directory to the Documents directory.

3.5 cd .. – Move One Level Up

The command `cd ..` moves the user one directory level upward from the current location.

Syntax:

```
cd ..
```

Example:

If the current directory is:

```
/home/user/Documents
```

After executing:

```
cd ..
```

the current directory becomes:

```
/home/user
```

3.6 cd – Change to Home Directory

When `cd` is used without any argument, it takes the user directly to their home directory.

Syntax:

```
cd
```

Example:

```
$ cd
```

The user is returned to their home directory.

3.7 cd ../../.. – Move Three Levels Up

The `..` represents the parent directory. Therefore, using it three times moves the user three directory levels upward.

Syntax:

```
cd ../../..
```

For example, if the current directory is:

```
/home/user/Documents/Linux/Lab
```

executing:

```
cd ../../..
```

moves the user three levels upward.

3.8 cd / – Move to Root Directory

The / represents the root directory of the Linux file system. The command `cd /` moves the user to the root directory.

Syntax:

```
cd /
```

Example:

```
$ cd /  
$ pwd  
/
```

3.9 mkdir <Foldername> – Create a Directory

The `mkdir` command is used to create a new directory.

Syntax:

```
mkdir <Foldername>
```

Example:

```
$ mkdir LinuxLab
```

This creates a new directory named `LinuxLab` in the current directory.

3.10 Creating a Directory Inside Another Directory

A directory can be created inside another directory by specifying the appropriate path.

Example:

`mkdir Foldername1/Foldername2`

For example:

`mkdir LinuxLab/Experiment1`

This creates Experiment1 inside the LinuxLab directory, provided that LinuxLab already exists.

3.11 rmdir <Foldername> – Remove an Empty Directory

The `rmdir` command removes a directory only when the directory is empty.

Syntax:

`rmdir <Foldername>`

Example:

`$ rmdir LinuxLab`

If LinuxLab contains files or subdirectories, the command will not remove it.

3.12 rm -r <Foldername> – Remove a Non-Empty Directory

The `rm -r` command is used to remove a directory along with its contents recursively.

Syntax:

`rm -r <Foldername>`

Example:

`$ rm -r LinuxLab`

This removes the LinuxLab directory and the files/subdirectories inside it.

Note: `rm -r` should be used carefully because deleted files and directories may not be recoverable through normal methods.

3.13 mv <Old Foldername> <New Foldername> – Rename a Directory

The `mv` command can be used to rename a directory.

Syntax:

`mv <Old Foldername> <New Foldername>`

Example:

```
$ mv LinuxLab LinuxBasics
```

This changes the directory name from LinuxLab to LinuxBasics.

Practical Implementation

The following sequence demonstrates several basic Linux operations:

```
$ pwd
$ ls
$ mkdir LinuxLab
$ ls -p
$ cd LinuxLab
$ pwd
$ mkdir Experiment1
$ ls -p
$ cd ..
$ mv LinuxLab LinuxBasics
$ ls -p
$ cd LinuxBasics
$ mkdir Test
$ cd ..
$ rmdir LinuxBasics
```

The last rmdir command will fail because LinuxBasics is not empty. To remove the directory and its contents, the following command can be used:

```
$ rm -r LinuxBasics
```

4. Discussion

In this experiment, several fundamental Linux commands were implemented to understand file-system navigation and directory management. The pwd command was used to identify the current working directory, while ls and ls -p were used to display directory contents.

The cd command provided practical experience in navigating between directories. The use of cd .., cd ../../.., and cd / demonstrated how Linux represents parent and root directories.

The mkdir command was used to create directories and subdirectories. On the other hand, rmdir was used to remove empty directories, while rm -r demonstrated how non-empty directories can be removed recursively. Finally, the mv command was used to rename directories.

These commands are essential for Linux system administration and provide a foundation for learning more advanced Linux commands, shell scripting, system administration, and cybersecurity.

5. Conclusion

The Linux Basics experiment provided practical knowledge of the Linux command-line interface and file-system management. By implementing commands such as `pwd`, `ls`, `cd`, `mkdir`, `rmdir`, `rm`, `-r`, and `mv`, the basic operations of navigating, creating, deleting, and renaming directories were successfully understood.

This experiment also helped develop familiarity with the hierarchical structure of the Linux file system. These basic commands are essential for working efficiently in Linux and serve as a foundation for more advanced Linux and system administration tasks.

6. References

1. Linux Manual Pages – `pwd`, `ls`, `cd`, `mkdir`, `rmdir`, `rm`, and `mv`.
2. William Shotts, *The Linux Command Line*, No Starch Press.
3. Linux Command-Line documentation and terminal help pages.